SEIZELUT: A LUT-NATIVE MULTIMODAL SEIZURE DETECTOR FOR WEARABLE DEVICES ON FPGA

Phase I: Dataset Study, Baselines and LUT Network Design

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by

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APPROVAL OF THE PROJECT EVALUATION BOARD

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DECLARATION

We certify that

a. The work contained in the report is original and has been done by us under the general supervision of our supervisor(s).

b. The work has not been submitted to any other Institute for any degree or diploma.

c. We have followed the guidelines provided by the Institute in writing the report.

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f. An AI assistant was used to help write analysis code, check results and draft parts of the text. All numbers in this report come from our saved experiment records and scripts, and we take responsibility for the content.

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Abstract

Automatic seizure detection with wearable sensors could give people with epilepsy objective seizure counts and timely alarms, but the detector has to run for days on a small battery, next to the sensors. In this project we ask whether a *LUT-native* neural network, whose neurons are the six-input lookup tables (LUTs) that make up an FPGA, can detect seizures from behind-the-ear EEG, ECG, EMG and motion signals in patients it has never seen, with no multipliers in the classifier and a budget of a few thousand LUTs.

We use SeizeIT2, a public wearable dataset of 125 patients with focal epilepsy (11,626 hours of recordings, 883 annotated seizures), with its official split into 96 training and 29 test patients. In Phase I, over the first two months of the project, we audited every file of the dataset and found that it differs from its documentation (all biopotentials are sampled at 256 Hz instead of 250 Hz, and the files hold 883 seizures instead of 886). We reproduced the published ChronoNet baseline: the published sensitivity was not reached (74.5 % against 84.2 %) at far fewer false alarms, and we found that the released code selects its model on the test patients, so we also trained a clean version that uses no test information. We designed a bit-exact integer feature pipeline of 117 features from the four modalities, with a fixed-point specification; it loses no accuracy against floating point (AUROC 0.736 against 0.735). We then compared nine model families with patient-wise five-fold cross-validation on the training patients and event-based SzCORE metrics.

On cross-validation, a Differentiable Weightless Neural Network of 50 six-input LUTs reaches an event-F1 of 0.211 with 2.4 false alarms per day, at an estimated 960 LUTs and no multipliers. It is seven to eight times smaller than the next LUT-based models (TreeLUT and a logic-gate network) and better on every metric than an INT8 neural network that needs 9,600 multiplications per decision, but it is still below gradient-boosted trees (event-F1 0.292). It detects 88 % of tonic-clonic seizures but few focal seizures, and it depends on the ECG signal. An interpretability study of the network showed that the first feature selection ignored the heart-rate features; selecting the inputs by LightGBM importance instead raised event-F1 from 0.148 to 0.208 at the same size, and training with random sensor dropout keeps the detector working when the ECG is missing (event-F1 0.172 against 0.092). The literature survey of 79 references found no LUT-native seizure detector and no LUT network that fuses modalities or

handles missing sensors.

Distillation from LightGBM was adopted as the final network design. With the design frozen, the models were evaluated once on the 29 held-out test patients with five seeds: the final network reached an event-F1 of 0.248 with 0.8 false alarms per day at about 900 LUTs, above the clean ChronoNet baseline (0.027 with 106 false alarms per day; per-patient $p = 0.027$) and TreeLUT (0.233), level with random forests (0.246) and below LightGBM (0.308). Its generated Verilog matches the software on all 431,960 windows of 10 test recordings. Phase II will implement and measure the detector on an FPGA.

Index Terms: epilepsy, seizure detection, wearable EEG, behind-the-ear EEG, multimodal fusion, weightless neural networks, lookup tables, FPGA, SeizeIT2.

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List of Abbreviations

ACC	Accelerometer
ASIC	Application-specific integrated circuit
AUPR	Area under the precision–recall curve
AUROC	Area under the receiver operating characteristic curve
bte-EEG	Behind-the-ear electroencephalography
CNN	Convolutional neural network
CV	Cross-validation
DSP	Digital signal processing block (FPGA multiplier)
DWN	Differentiable Weightless Neural Network
ECG	Electrocardiography
EDF	European Data Format
EFD	Extended Finite Difference
EMG	Electromyography
FA	Focal aware (seizure); false alarm
FBTC	Focal to bilateral tonic-clonic (seizure)
FIA	Focal impaired awareness (seizure)
FP/24 h	False positives (false alarms) per 24 hours
FPGA	Field-programmable gate array
GBDT	Gradient-boosted decision trees
GRU	Gated recurrent unit
GYR	Gyroscope
iEEG	Intracranial EEG
ILAE	International League Against Epilepsy
INT8	8-bit integer format
LGN	Logic-gate network
LSB	Least significant bit

LUT	Lookup table (LUT6: six-input lookup table)
MCU	Microcontroller
MI	Mutual information
MIL	Multiple-instance learning
MLP	Multilayer perceptron
OVLP	Any-overlap event scoring
SoC	System on chip
SUDEP	Sudden unexpected death in epilepsy
SVM	Support vector machine
SzCORE	Seizure Community Open-Source Research Evaluation framework
TCN	Temporal convolutional network
TUSZ	Temple University Hospital Seizure Corpus
WNN	Weightless neural network

Chapter 1

Introduction

1.1 Background and Motivation

Epilepsy is a common neurological disorder: an estimated 45.9 million people had active epilepsy in 2016 [1]. Seizures are classified by onset; focal seizures start in one part of the brain, may or may not impair awareness, and can spread into bilateral tonic-clonic seizures [2]. Patients are often unaware of their own seizures, so seizure diaries undercount them [3], and tonic-clonic seizures carry a risk of sudden unexpected death in epilepsy [4]. Reliable, automatic seizure detection outside the hospital would give clinicians objective seizure counts and caregivers timely alarms.

In hospital, seizures are diagnosed with video-EEG from a full scalp electrode cap, which cannot be worn at home for days. Wearable devices are recommended today only for detecting tonic-clonic seizures [5]; for focal seizures that do not become tonic-clonic (822 of the 883 seizures in SeizeIT2) the problem is open, and false alarms are the main obstacle [6]. Behind-the-ear EEG combined with ECG, EMG and motion sensors is a discreet alternative [7, 8], and the SeizeIT2 dataset [9] makes it possible to develop and evaluate detectors on 125 patients recorded with such a wearable.

A wearable detector has to run for days on a small battery, close to the sensors, and raise an alarm within seconds. Sending raw signals to a phone or a server costs radio energy and depends on a connection; running a neural network on a microcontroller or an accelerator costs multiplications, which dominate the area and power of digital hardware. Field-programmable gate arrays (FPGAs) offer a different trade-off: their basic logic element is a small lookup table (LUT), typically with six inputs, that implements any Boolean function of those inputs in one clock cycle.

A family of recent machine-learning models, which we call *LUT-native*, is built from exactly this element: weightless neural networks, whose neurons are truth tables [10], logic-gate networks [11], and LUT-based networks designed for FPGAs [12]. They need no multipliers

at inference time, so a classifier can cost a few hundred to a few thousand LUTs. They have been applied to image benchmarks, physics triggers and, very recently, ECG [13] and EEG [14] classification, but, to our knowledge, not to seizure detection on multimodal wearable data.

1.2 Problem Statement and Objectives

> *Can a LUT-native neural network, fed by integer features computed from behind-the-ear EEG, ECG, EMG and motion, detect seizures in patients it has never seen as well as the published SeizeIT2 baselines, with no multipliers in the classifier and a budget of a few thousand FPGA LUTs?*

The objectives of the project are:

1. Audit the SeizeIT2 dataset [9] completely and reproduce its published baselines.
2. Design an integer-only feature pipeline that a hardware front end can compute bit-exactly, and show that it loses no accuracy against floating point.
3. Compare LUT-native models (DWN, logic-gate networks, TreeLUT, LogicNets-style networks) with software references (gradient-boosted trees) and multiplier-based networks (INT8 MLP, CNN/TCN, GRU) on the same features, splits and metrics, reporting accuracy against estimated hardware cost.
4. Extend the best LUT-native model with ideas specific to wearable seizure detection (memory across seconds, cheaper input encoders, adaptation to new patients, learning from a stronger model, losses for rare and noisily labelled events, robustness to missing sensors, interpretability), keeping only what is statistically significant.
5. Evaluate the final models once on the 29 held-out test patients with event-based metrics (SzCORE [15]).
6. Deliver a hardware handoff package (truth tables, thresholds, test vectors, verified reference Verilog) and implement the detector on an FPGA.

Objectives 1–5 and the hardware handoff of objective 6 were completed in Phase I and are reported here; the FPGA implementation of objective 6 is the work of Phase II.

1.3 Expected Contributions

- The first differentiable LUT-native seizure detector, and the first benchmark of LUT-family models on SeizeIT2.

- A bit-exact integer multimodal feature front end with its fixed-point specification.
- An independent reproduction of the SeizeIT2 ChronoNet baseline, including a protocol that uses no test information.
- Extensions of the DWN for biosignal events (stateful LUT networks, comparator-budgeted encoders, robustness to missing sensors), each judged by per-patient statistics.
- A hardware handoff package that lets the FPGA implementation be checked bit for bit against the software.

1.4 Organisation of the Report

Chapter 2 reviews the literature on wearable seizure detection, detection algorithms and their evaluation, LUT-native networks and hardware seizure detectors, and identifies the research gaps. Chapter 3 describes the dataset and our audit of it, the proposed system, the models compared and the evaluation protocol. Chapter 4 presents the experimental results of Phase I. Chapter 5 describes the problems we met and how we solved them. Chapter 6 discusses the findings, summarises the progress and gives the plan for Phase II. Appendix A lists the source file of every reported number.

Chapter 2

Literature Review

This chapter reviews the literature in five areas, from the clinical problem to the hardware: epilepsy and the need for automatic detection (Section 2.1), wearable and multimodal detection (Section 2.2), detection algorithms and their evaluation (Section 2.3), LUT-native, weightless and logic networks (Section 2.4), and hardware seizure detectors (Section 2.5). Section 2.6 summarises the research gaps. Every reference was checked against a published record (title, authors, venue and year); numbers quoted from a paper are as reported in its abstract unless stated otherwise.

2.1 Epilepsy and the Need for Automatic Seizure Detection

Epilepsy affected an estimated 45.9 million people with active epilepsy in 2016 [1]. The ILAE 2017 operational classification [2] divides seizures by onset into focal and generalised; focal seizures are further described by whether the patient stays aware (focal aware, focal impaired awareness), and a focal seizure can spread into a focal to bilateral tonic-clonic seizure. These are the classes annotated in SeizeIT2.

Patients are often unaware of their own seizures, which Elger and Hoppe identify as the main cause of seizure under-reporting [3]; seizure diaries therefore undercount, and treatment decisions are made on incomplete information. Seizures are also dangerous: in the MORTEMUS study of sudden unexpected death in epilepsy (SUDEP) in monitoring units, the fatal cardiorespiratory dysfunction followed a generalised tonic-clonic seizure [4]. Objective, automatic seizure counting outside the hospital is therefore a clinical need.

The ILAE–IFCN clinical practice guideline [5] gives a weak recommendation for wearable devices that detect generalised and focal to bilateral tonic-clonic seizures, and does not recommend current devices for other seizure types. Detecting *focal* seizures, the majority in focal epilepsy, with a wearable remains open.

2.2 Wearable and Multimodal Seizure Detection

Reviews and devices. A review of multimodal wearable detection [16] found sensitivities from 4 % to 100 % across studies, with tonic-clonic seizures detected best. A wearable EMG device detected tonic-clonic seizures with more than 90 % sensitivity in a 71-patient study [17], and wrist-worn electrodermal-activity and accelerometer sensors reached 94.55 % sensitivity at 0.2 false alarms per day for convulsive seizures [18]. A meta-analysis [19] pooled a sensitivity of 0.91 for tonic-clonic seizures at 2.1 false alarms per 24 hours. All of this evidence concerns convulsive seizures.

Behind-the-ear EEG and the SeizeIT project. EEG is the most direct signal of a seizure, but a full scalp EEG is not wearable for days. The SeizeIT project records EEG from electrodes behind the ears. Vandecasteele et al. [7] showed that seizures can be annotated and detected on behind-the-ear EEG, with a patient-specific detector at 69.1 % sensitivity and 0.49 false positives per 24 h and a patient-independent one at 64.1 % and 2.8 per 24 h. Adding ECG to a two-channel EEG detector raised sensitivity by 8–11 % at the same false-alarm rate [8], and helped human reviewers too: with ECG, false detections fell from 6.5 to 2.4 per day [20]. The ICASSP 2023 Grand Challenge on wearable-EEG seizure detection [21] was won by an ensemble of decision trees. SeizeIT2 [9] then released 125 patients with more than 11,000 hours of behind-the-ear EEG, ECG, EMG and motion, with ChronoNet and SVM baselines; it is the dataset of this project. A multicentre clinical validation of the same wearable on 192 adults [6] reached an algorithmic sensitivity of 0.73 at a precision of only 0.004 before human review, which shows that false alarms are the main obstacle for focal seizures.

Other work on SeizeIT2. A study of tonic-clonic seizures in the SeizeIT2 cohort [22] fused EEG, EMG and accelerometry and reached 90.9 % sensitivity at 0.1 false positives per 24 h, against 100 % at 10.3 per 24 h for EEG alone: fusion mainly removes false alarms. ECG-only anomaly detection [23] and a complex-valued multimodal network [24] have also been applied to SeizeIT2; none of these reports a hardware budget.

Fusion and missing modalities. BrainFuseNet [25] fused EEG, PPG and accelerometry on a microcontroller (64.28 % sensitivity at 0.21 false positives per hour). A cyclic transformer trained on EEG and EMG but tested on EEG alone [26] is the only work we found that targets a missing modality at test time. Table 2.1 summarises this section.

Table 2.1 Wearable and multimodal seizure detection studies (results as reported).

Work	Year	Signals	Seizures	Result	Gap for our problem
Beniczky et al. [17]	2018	EMG	tonic-clonic	>90 % sensitivity	convulsive only
Onorati et al. [18]	2017	EDA + ACC	convulsive	94.55 %, 0.2 FA/day	convulsive only
Naganur et al. [19]	2022	meta-analysis	tonic-clonic	0.91, 2.1 FA/24 h	focal not solved
Vandecasteele et al. [7]	2020	bte-EEG	focal	64.1 %, 2.8 FP/24 h (patient-independent)	EEG only
Vandecasteele et al. [8]	2021	bte-EEG + ECG	focal	+8–11 % sensitivity with ECG	software only
Bhagubai et al. [21]	2024	bte-EEG	challenge	tree ensemble won	no hardware budget
Zhang et al. [22]	2024	EEG + EMG + ACC	tonic-clonic	90.9 %, 0.1 FP/24 h	tonic-clonic only
Swinnen et al. [6]	2026	SeizeIT2 wearable	focal, 192 adults	0.73 at precision 0.004	false alarms
BrainFuseNet [25]	2024	EEG + PPG + ACC	–	64.28 %, 0.21 FP/h, MCU	multipliers, no ECG/EMG
Zhang et al. [26]	2024	EEG + EMG	–	–	large transformer

2.3 Seizure Detection Algorithms and Their Evaluation

Feature-based detectors. Shoeb and Guttag [27] trained a patient-specific support vector machine on EEG spectral features and detected 96 % of 173 test seizures on the CHB-MIT scalp EEG database, with a median latency of 3 s and 2 false detections per 24 h. Patient-specific training needs seizures of the same patient, which a wearable for new patients does not have. On wearable behind-the-ear EEG, the winner of the ICASSP 2023 Grand Challenge, SeizFt [28], used hand-crafted features, Fourier-transform-surrogate augmentation and a gradient-boosted tree ensemble (CatBoost); a random forest on band-power features reached 73.6 % sensitivity and 96.7 % specificity in the same challenge [29]. Tree ensembles are the strongest published approach on SeizeIT data, which is why LightGBM, XGBoost and CatBoost are our software references.

Deep-learning detectors. ChronoNet [30] stacks multi-scale one-dimensional convolutions and densely connected GRU layers on raw EEG; it was designed for abnormal-EEG classification and is the deep baseline of SeizeIT2. In the ICASSP 2023 challenge, ChronoNet trained on SeizeIT1 reached only 15.2 % sensitivity on the SeizeIT2 validation patients [29]. EEGNet [31] is a compact convolutional network for EEG; EEGWaveNet [32] extracts multiscale features with trainable depthwise convolutions and improves subject-independent detection; graph neural networks with self-supervised pre-training [33] reached a detection AUROC of 0.875 on TUSZ but need a full electrode montage to build the graph; U-Net transformers [34] and multi-signal transformers such as BIOT [35] push accuracy further with much larger models. These networks are floating-point and multiplier-based; their size is reported in parameters, not in hardware cost.

Learning from imperfect labels. Seizure onset and offset are uncertain even for experts. Weak supervision with noisy clinical labels [36], attention-based multiple-instance learning when only time periods are labelled [37], and uncertainty-aware training with noisy labels [38] address this on scalp EEG. Knowledge distillation from a full-montage teacher into a two-channel student [39] keeps detection competitive with very few channels. These ideas motivate our extensions N4 (distillation) and N5 (noise-robust and multiple-instance losses), applied here to a LUT network.

Datasets and evaluation. Most detectors are developed on scalp EEG: CHB-MIT and the TUH Seizure Corpus [40]. How a detector is scored changes its numbers: epoch-based scoring, any-overlap event scoring and time-aligned event scoring give different results for the same system [41]. SzCORE [15] standardised data formats, cross-validation and event-based metrics (sensitivity, precision, F1, false positives per day) with an open-source scorer. In the first blind

SzCORE challenge [42], on 65 unseen subjects and 4,360 hours of EEG, the best algorithm reached an event-F1 of 43 % (sensitivity 37 %, precision 45 %), and reported results did not hold up on blind data. Data leakage, letting a subject's data appear in both training and test sets, is common in EEG deep learning and inflates results [43]. Table 2.2 summarises this section.

Table 2.2 Seizure detection algorithms and evaluation frameworks (results as reported).

Work	Year	Method	Data	Result	Gap for our problem
Shoeb & Guttag [27]	2010	patient-specific SVM	CHB-MIT	96 % of seizures, 2 FD/24 h	patient-specific, scalp
SeizFt [28]	2023	features + CatBoost	SeizeIT1/2	ICASSP 2023 winner	EEG only, no hardware cost
Huang et al. [29]	2023	random forest; ChronoNet	SeizeIT1/2	RF 73.6 %; ChronoNet 15.2 %	EEG only
ChronoNet [30]	2019	CNN + GRU	TUH abnormal	benchmark at the time	large, multipliers
EEGWaveNet [32]	2022	multiscale CNN	CHB-MIT, TUSZ	better subject-independent	scalp, float CNN
Tang et al. [33]	2022	self-supervised GNN	TUSZ	AUROC 0.875	needs full montage
Shama & Venkataraman [38]	2026	noisy-label training	TUH, CHB-MIT	robust to label noise	scalp EEG
Zheng et al. [39]	2023	distilled GNN	TUSZ	competitive with 2 channels	float student
SzCORE [15]	2024	evaluation framework	scalp EEG	standard event metrics	not for wearables
SzCORE challenge [42]	2025	blind benchmark	65 subjects	best event-F1 43 %	scalp EEG

2.4 LUT-Native, Weightless and Logic Networks

Weightless neural networks. A weightless network replaces weighted sums with memory lookups. WiSARD [44] split a binary input into small tuples, each addressing a RAM node, with one discriminator per class whose score is the number of RAM nodes that fire. Such networks are trained by writing tables, not by gradient descent, and their inputs must already be binary. BTHOWeN [45] added Bloom filters, hashing and a Gaussian thermometer encoding of continuous inputs, with an FPGA accelerator; ULEEN [46] borrowed training techniques from binarised networks and reported 3.6 times the area-normalised throughput and 7.1 times the energy efficiency of a FINN binarised network on FPGA.

Differentiable Weightless Neural Networks. The DWN [10] made the lookup tables themselves trainable end to end: thermometer-encoded inputs, LUT layers trained with an Extended Finite Difference gradient, and a learnable input mapping that decides which bits each LUT reads. The authors report better latency, throughput, energy and area than earlier FPGA designs, and better accuracy than XGBoost on a microcontroller under tight memory limits. DWNs have since been applied to wearable inertial activity recognition [47] (96.3–96.7 % accuracy at 56–104 nJ per sample) and to continuous control with thermometer-encoded sensor inputs compiled to FPGA logic [48]. The DWN is the base model of this project.

Logic-gate networks. Deep Differentiable Logic Gate Networks [11] learn, for every neuron, which of the 16 two-input Boolean functions to compute, and after training run as pure logic (over a million MNIST images per second on one CPU core). Convolutional LGNs [49] reach about 86 % on CIFAR-10 with tens of millions of gates; Light LGNs [50] reduce the parameters of higher-fan-in gates; a recurrent LGN [51] is the only stateful logic network we found, evaluated on machine translation.

LUT-based networks for FPGAs. LUTNet [52] replaced the XNOR operations of binarised networks with trainable K-input LUTs. LogicNets [12] trains sparse networks with few-bit activations so that every neuron can be enumerated as a truth table and synthesised directly into FPGA logic, with latencies below 15 ns; its cost grows exponentially with fan-in, which keeps fan-in tiny. PolyLUT [53] lets each LUT neuron compute a polynomial of its inputs, and NeuraLUT [54] hides a small dense sub-network inside each LUT (up to 4.3 times lower latency at equal accuracy). AmigoLUT [55] scales LUT networks through ensembles, and ReducedLUT [56] compresses the tables by treating input patterns never seen in training as "don't care" (up to 1.63 times fewer LUTs), an assumption that is risky when a new patient produces patterns the training patients did not. TreeLUT [57] maps quantised gradient-boosted trees into LUT-only logic with 4–101 times lower area-delay product than recent LUT networks; we use it as a strong non-neural LUT baseline.

The cost of the encoder. Mecik and Kumm [58] built a complete DWN hardware generator and found that the thermometer encoder can raise LUT usage by up to 3.2 times and dominates the cost of small networks. Our own cost estimate agrees: in our 50-LUT base model the encoder accounts for about 850 of about 960 LUTs (Chapter 3), which is why encoder design (N2) is one of our extensions.

LUT and logic networks on biosignals. Mommen et al. [13] classified ECG arrhythmias across patients with LGNs and LUT networks, up to 94.3 % accuracy on MIT-BIH with 2,000–2,990 LUTs on an Artix-7 FPGA (about 5–7 mW, estimated). Dharia et al. [14] applied LGNs to

EEG classification (dementia and emotion) on an embedded CPU. The only weightless network applied to seizures is a WiSARD model for seizure anticipation evaluated on 4 patients, with accuracy as the only metric [59]. Table 2.3 summarises this section.

Table 2.3 LUT-native, weightless and logic networks (results as reported).

Work	Year	Model	Domain	Hardware / result	Gap
WiSARD [44]	1984	RAM-node WNN	images	–	no gradient training
ULEEN [46]	2023	weightless	MLPerf Tiny	7.1× energy eff. vs FINN	no biosignals
DWN [10]	2024	differentiable LUT6 net	tabular	FPGA, MCU	stateless, no events
nanoML HAR [47]	2025	DWN	inertial HAR	56–104 nJ / sample	single modality
LGN [11]	2022	2-input logic gates	images	>1 M images/s (CPU)	no biosignals
Recurrent LGN [51]	2025	stateful logic gates	translation	–	no hardware
LogicNets [12]	2020	sparse quantised LUT net	jet tagging	<15 ns latency	tiny fan-in
NeuraLUT [54]	2024	dense sub-nets in LUTs	jet tagging	4.3× lower latency	static
TreeLUT [57]	2025	quantised GBDT	tabular	4–101× lower area-delay	per window
Mecik & Kumm [58]	2025	DWN generator	jet tagging	encoder up to 3.2× LUTs	–
Mommen et al. [13]	2026	LGN, LUT network	ECG arrhythmia	2.0–3.0k LUTs, 5–7 mW	single modality
Dharia et al. [14]	2026	LGN	EEG	embedded CPU	no FPGA, no seizures
de Aguiar et al. [59]	2015	WiSARD	seizure anticipation	–	4 patients, accuracy only

2.5 Hardware Seizure Detectors and Efficient Inference

ASICs. The first micro-power seizure-detection chips put the analogue front end and feature extraction on chip: Verma et al. [60] extracted spectral EEG features at 9 µJ per feature vector and sent them to a central classifier, which cut system power 14 times by avoiding raw-EEG transmission. Later systems-on-chip ran the classifier too: a 16-channel patient-specific SVM chip at 2.73 µJ per classification [61], a non-linear SVM chip at 1.83 µJ per classification [62], and an SVM processor with on-chip learning at 1.9 mW [63]. Tree-based classifiers proved

particularly cheap: gradient-boosted trees on intracranial EEG at 41.2 nJ per classification [64], and the NeuralTree chip at 0.227 µJ per classification across 256 channels [65]. Trees only compare and add, which is also why they map well to lookup tables.

FPGAs. FPGA detectors include an STFT and SVM design on a Zynq-7000 at 380 mW of programmable-logic power [66], SVM training and classification on a Virtex-7 [67], a patient-specific intracranial-EEG detector on a Cyclone V at 0.49 mW dynamic power [68], an ultra-low-complexity line-length detector [69], and a spiking neural network on a small Lattice iCE40UP5K [70], which reported 4.55 nJ and 0.5 µs per inference with 0.3 false positives per hour on CHB-MIT using four temporal EEG channels. The last is the closest to our Phase II target: a low-cost FPGA and few channels near the ear.

Microcontrollers. e-Glass [71] put four EEG electrodes in a glasses frame and ran detection on an embedded processor for 2.71 days per charge. EEGformer [72] ran a compact transformer on temporal EEG channels at 0.31 mJ per inference and detected 73 % of CHB-MIT seizures at 0.15 false positives per hour. BrainFuseNet [25] fused EEG, PPG and accelerometry on the same class of microcontroller at 0.11 mJ per inference. Tree-based EEG artefact detection on a low-power processor [73] and the BioWolf platform [74] show the same push towards wearable, always-on processing.

Tools for efficient inference. Binarised neural networks [75] reduce weights and activations to one bit; FINN [76] compiles them into streaming FPGA dataflow; integer-only quantisation [77] made 8-bit inference standard; hls4ml [78] translates trained networks into FPGA firmware with latencies around 100 ns. Quantised networks still need integer multipliers (FPGA DSP blocks); our INT8 MLP baseline represents this approach. Table 2.4 summarises the hardware detectors.

2.6 Research Gaps

Table 2.5 lists the gaps found in the literature and how this project addresses each of them.

Table 2.4 Hardware seizure detectors (results as reported; units differ between papers).

Work	Year	Platform	Signals	Hardware	Detection	Gap
Verma et al. [60]	2010	ASIC	scalp EEG	9 µJ / feature vector	classifier off-chip	EEG only
Bin Altaf et al. [61]	2015	ASIC	scalp EEG	2.73 µJ / class.	95.7 % (epoch)	per patient
Shoaran et al. [64]	2018	ASIC	iEEG	41.2 nJ / class.	F1 87.9 %	implant
NeuralTree [65]	2022	ASIC	EEG, iEEG	0.227 µJ / class.	95.6 % (EEG)	neural only
Wang et al. [66]	2018	FPGA (Zynq)	scalp EEG	380 mW	98.4 %, 0.356 FD/h	per patient
Farhang Razi & Schmid [68]	2022	FPGA (Cyclone V)	iEEG	0.49 mW dynamic	100 % sens.	implant
Busia et al. [70]	2025	FPGA (iCE40)	scalp EEG	4.55 nJ / inference	0.3 FP/h	EEG only
e-Glass [71]	2018	MCU	scalp EEG	2.71 days / charge	93.8 % (epoch)	EEG only
EEGformer [72]	2024	MCU	scalp EEG	0.31 mJ / inference	73 %, 0.15 FP/h	multipliers
BrainFuseNet [25]	2024	MCU	EEG, PPG, ACC	0.11 mJ / inference	0.21 FP/h	no ECG/EMG

Table 2.5 Research gaps and how this project addresses them.

#	Gap	Evidence	How this project addresses it
1	No LUT-native (weightless, logic-gate or LUT) seizure detector exists	LUT networks are benchmarked on images, physics and tabular data; the only weightless seizure work is a 4-patient study [59]	First benchmark of DWN, LGN, TreeLUT and LogicNets-style networks for seizure detection, on SeizeIT2
2	Hardware detectors use EEG alone, although fusion is the most consistent way to cut false alarms	[8, 22]; Section 2.5	Integer features from behind-the-ear EEG, ECG, EMG and motion in one bit-exact front end
3	Missing sensors are rarely handled; LUT networks never	[26]; Section 2.4	Presence bits, modality dropout, gated and late fusion, missing-sensor tests (N6)
4	The encoder's hardware cost is ignored, although it dominates small LUT networks	[58]; our estimate: about 850 of 960 LUTs	Cost model that counts every comparator; comparator-budgeted and learned encoders (N2)

#	Gap	Evidence	How this project addresses it
5	LUT networks are stateless; seizures unfold over tens of seconds	[51]	A stateful DWN with state bits fed back every second (N1)
6	Results are patient-specific, leak between training and test, or use incomparable metrics	[41–43]	Patient-wise cross-validation, a locked test set used once, SzCORE metrics with the paper's metrics alongside
7	Small models learn from noisy labels and from larger models only in floating point	[38, 39]	Noise-robust and multiple-instance losses (N5) and distillation from LightGBM (N4) into a LUT network
8	Detectors are black boxes	Section 2.4	Tracing alarms to the inputs read by each LUT (N7)
9	Published wearable baselines are not independently reproduced	Section 2.2	Independent reproduction of the SeizeIT2 ChronoNet baseline, with and without test-patient information

Chapter 3

Dataset, Methodology and Experimental Setup

3.1 SeizeIT2 Dataset

We use SeizeIT2 [9], a public multimodal wearable dataset of patients with focal epilepsy, released on OpenNeuro as ds005873 (version 1.1.0, licence CC0). The patients were monitored in the epilepsy monitoring units of five European centres (Leuven, Coimbra, Freiburg, Karolinska and Aachen) with a wearable device while the hospital video-EEG was recorded for the clinical annotation of every seizure. The wearable records four modalities (Table 3.1).

Table 3.1 Signals recorded by the SeizeIT2 wearable (sampling rates measured from the files).

Modality	**Module (from the channel labels)**	**Channels**	**Rate**
EEG	EEG module, electrodes behind the ears	2: BTEleft, BTEright or a cross-head channel	256 Hz
ECG	ECG/EMG module	1	256 Hz
EMG	ECG/EMG module	1	256 Hz
Motion	ECG/EMG module and EEG module	3-axis accelerometer + 3-axis gyroscope per module	25 Hz

The dataset defines the split we follow: **96 training patients (sub-001 to sub-096) and 29 test patients (sub-097 to sub-125)**, balanced by the authors for seizure count, data amount, centre and seizure type. All model development uses only the 96 training patients, with five-fold patient-wise cross-validation inside them; the 29 test patients are scored once, after every design choice is frozen.

3.2 Data Audit

Before building any model we measured every file of the release (11,009 EDF files, 125.9 GB) instead of relying on the documentation, because a hardware pipeline has to be designed for the exact sampling rates, signal ranges and missing channels. The audit streams each file from the public S3 bucket, measures it and deletes it, so it needs no local copy of the dataset. Table 3.2 gives the inventory and Table 3.3 the seizure classes.

Table 3.2 Inventory of SeizeIT2 measured by our audit.

Quantity	Value
Patients	125 (train 96, test 29)
Recordings (runs)	2,850
Files per modality	EEG 2,850; ECG 2,804; EMG 2,804; motion 2,551
Hours of EEG	11,626 h (median 95 h per patient, range 1.1–397 h)
Seizures	883 (train 702, test 181)
Seizure time	14.3 h, i.e. 0.123 % of the recording time
Seizure duration	2 s to 962 s, median 33 s; 87 seizures shorter than 10 s
Missing ECG / EMG	46 runs (3.1 % of the time)
Missing motion	299 runs (3.9 % of the time)

Table 3.3 Seizure classes in the training and test partitions.

Seizure class	Train	Test	Total
Focal impaired awareness (FIA)	289	102	391
Focal aware (FA)	274	42	316
Focal, awareness unknown	105	10	115
Focal to bilateral tonic-clonic (FBTC)	33	22	55
Unknown onset	1	5	6
All	702	181	883

Two facts shape the whole project. First, seizures are rare: 0.123 % of the time, so a detector that never fires is 99.88 % “accurate” and accuracy is useless as a metric (Section 3.8). Second, most seizures are focal (only 55 of 883 are tonic-clonic), and focal seizures are the hardest to see on two behind-the-ear electrodes.

3.2.1 Differences between the files and the documentation

The audit found several differences between the released files and the documentation (Table 3.4). Each changed a design decision.

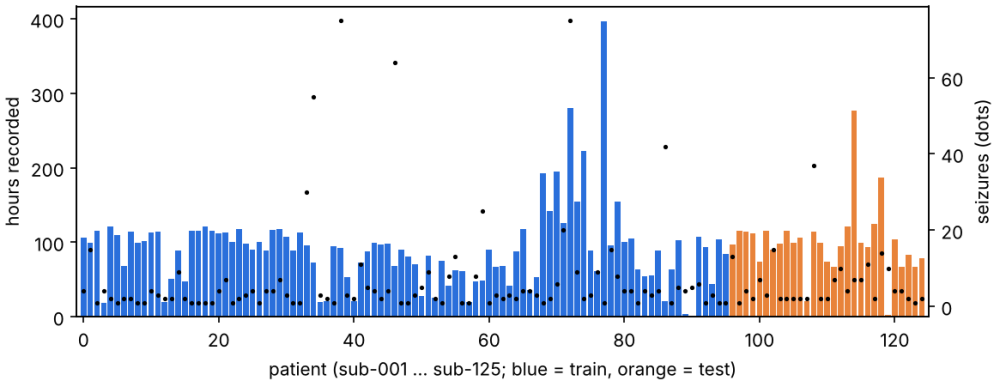


Figure 3.1 Hours recorded per patient (bars; blue: training, orange: test) and number of seizures per patient (dots).

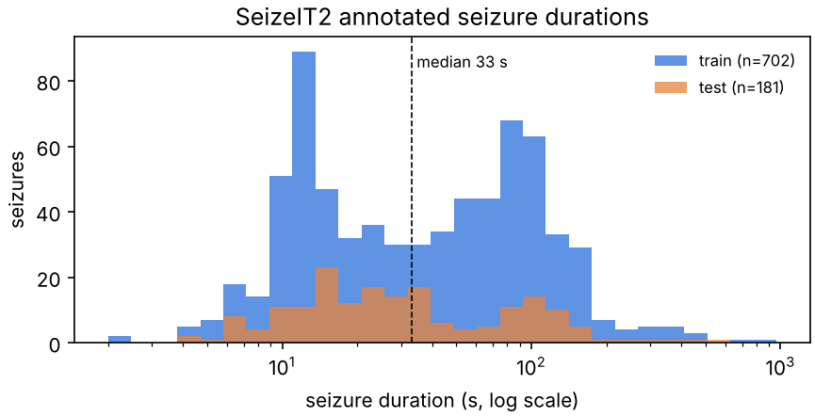


Figure 3.2 Distribution of annotated seizure durations (logarithmic axis).

Table 3.4 Where the released files differ from the documentation.

Item	Documentation	Files	What we did
EEG/ECG/EMG sampling rate	250 Hz	256 Hz in every file	Windows defined in seconds; integer filters designed for 256 Hz
Number of seizures	886 (train 704, test 182)	883 (train 702, test 181)	Report our count and cite the paper's
Seizure classes	includes "subclinical"	no such class	Breakdowns use the classes present
Shortest seizure	3 s	2 s	Kept
Female patients	51	62 in the participants file	Reported as a discrepancy
Centre and age per patient	used for stratification	not in the public release	Folds stratified by seizure count only
Seizure visibility on the wearable	expected	not annotated	Visible-seizure analysis dropped
Gaps inside files	"discontinuous"	every file continuous	A window never crosses a file boundary

3.2.2 Signal quality

The official baseline discards windows whose RMS is outside 13–150 µV. On behind-the-ear channels the typical RMS is close to the 13 µV floor (median per-run RMS 8.0 µV and 12.1 µV for the two channels), so the rule flags 69 % and 47 % of ordinary windows. We also found that the released evaluation code combines the two channels with a Python `and` on two lists, which returns only the second list, so in practice only one channel is gated. We reproduce this behaviour for the baseline and, for our own models, feed the artefact flags as input bits instead of gating blindly. Motion sensors are exactly constant for 22.8 % of channel time (sensor idle), so a flat motion signal does not mean a missing file; presence bits come from file availability.

3.3 Design Goals

The detector is meant to run on a small FPGA inside a wearable, next to the sensors. This fixes five requirements:

1. **Integer arithmetic only**, from the raw samples to the alarm, so that the software model and the hardware give bit-identical results.
2. **No multipliers in the classifier.** Multipliers (FPGA DSP blocks) dominate area and power in neural-network accelerators; a LUT network uses the FPGA's native logic element instead.
3. **A budget of a few thousand LUTs** for the classifier, small enough for a low-power FPGA.
4. **Generalisation to new patients:** trained once on other patients, it must work on a new patient without recalibration.
5. **Missing sensors are normal:** about 3–4 % of the recording time has no ECG/EMG or motion.

3.4 System Architecture

Figure 3.3 shows the proposed detector. Every second the front end produces one feature vector describing the last 2 seconds (2-s windows, 1-s hop). The encoder turns each selected feature into 8 bits by comparing it with 8 thresholds; the LUT network maps those bits to an integer score; the decision logic raises an alarm when enough recent seconds are positive.

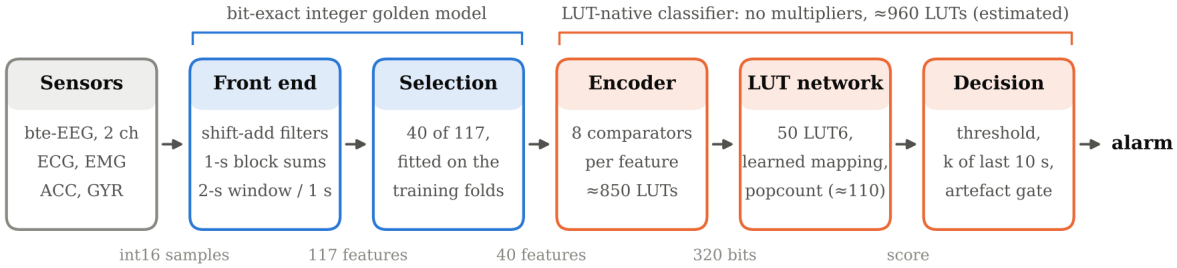


Figure 3.3 Block diagram of the proposed detector, with the estimated LUTs of the base configuration. Blue blocks form the bit-exact integer front end; orange blocks form the LUT-native classifier.

3.5 Integer Feature Extraction

The front end is written as a *golden model*: a bit-exact software description of the hardware, with a fixed-point specification of every signal width, scale and rounding rule. Inputs are 16-bit integers (0.25 µV per bit for biopotentials). Filters are second-order sections whose coefficients are sums of powers of two, so that each "multiplication" is a few shifts and additions; error feedback keeps the rounding noise below one bit. Windows are built from two 1-s block accumulators, so means are shifts. Logarithms are the position of the leading one plus three mantissa bits. Table 3.5 lists the 117 features.

Table 3.5 The 117 integer features computed for every 2-s window.

Group	Count	Features
EEG	84	per channel and across channels: mean absolute value, line length, mobility, zero crossings, extrema count, band amplitude in the delta, theta, alpha, beta and high-frequency bands; channel sign agreement; artefact flags
ECG	5	$\log_2$ of the RR interval (heart rate), $\log_2$ of the mean absolute successive RR difference, signal-quality flag
EMG	8	mean absolute value, line length, zero crossings, high/low frequency ratio
Motion	16	accelerometer and gyroscope magnitude, jerk, activity count, band amplitude
Presence	4	one bit each for ECG, EMG, accelerometer and gyroscope being available

Many features also have a *context* version: the difference between the feature and a causal 64-s exponential moving average of itself, so the network sees the change from the patient's recent baseline (for example, heart rate rising). Heart rate is computed from integer R-peak timestamps without any division. A floating-point twin of the same pipeline measures what the integer arithmetic costs.

3.6 The LUT-Native Classifier: DWN

A weightless neural network has no weights to multiply. Each neuron is a truth table: it reads a few input bits, forms an address and outputs the bit stored at that address. On an FPGA a six-input truth table is exactly one LUT6, so one neuron costs one LUT. The Differentiable Weightless Neural Network (DWN) [10] makes such networks trainable by gradient descent:

- **Thermometer encoding.** Feature x_f becomes the bits

$$b_{f,i} = [x_f > t_{f,i}], \qquad i = 1, \ldots, 8, \tag{3.1}$$

  with thresholds $t_{f,1} < \cdots < t_{f,8}$ placed at quantiles of the training data. In hardware each bit is one comparator against a constant.

- **LUT layer with learnable mapping.** Each of the L neurons reads $n = 6$ bits. Which bits it reads is learned during training; the 64 table entries are learned with the Extended Finite Difference gradient, and a straight-through estimator passes gradients between layers.

- **Popcount score.** The neurons are split into a seizure group S and a background group B, and the score is the integer

$$s = \sum_{j \in S} y_j - \sum_{j \in B} y_j, \tag{3.2}$$

  where $y_j \in \{0, 1\}$ is the output of neuron j; it is computed with an adder tree.

Our base configuration is $L = 50$ LUT6 neurons in one layer, 8 thresholds per feature, learnable mapping, and the 40 most useful features chosen inside each training set. Its estimated cost is about 960 LUTs, of which the thermometer encoder is about 850 and the network itself 50: the encoder, not the network, dominates. We re-implemented the DWN in PyTorch so that it trains on CPUs as well as GPUs; unit tests check that it reproduces the reference CUDA kernel exactly.

Feature selection. The 40 inputs are chosen inside each training set only, never using validation patients: first by mutual information and, after the interpretability study (Chapter 4), by the gain importance of a LightGBM model trained on the same training windows.

3.7 Models Compared

Table 3.6 lists the model families, and Table 3.7 the extensions of the DWN studied in this project.

Table 3.6 Models compared on the same features, splits and metrics.

Group	Models	Role
Published baselines	ChronoNet [30] (CNN + GRU on raw EEG); SVM on 42 EEG features [7]	reference results of the SeizeIT2 paper
Software references	logistic regression, random forest, XGBoost, LightGBM, CatBoost	best accuracy reachable on our features
Hardware-oriented models	INT8 MLP (multipliers), TreeLUT [57], logic-gate network [11], LogicNets-style network [12], DWN [10]	accuracy per LUT
Other references	GRU on features; 1-D CNN and TCN on raw EEG	sequence and raw-signal references

Table 3.7 Extensions of the DWN (N1–N7). Each is adopted only if it is significantly better than the base.

ID	Idea	Question
N1	Stateful LUT network: 16 state bits fed back every second; stacked past windows	Does memory of the last minute help?
N2	Encoder design: uniform, Gaussian or learned thresholds under a comparator budget	Can the encoder, the largest cost, be cheaper or better?
N3	Label-free per-patient threshold adaptation	Can the detector adapt to a new patient without labels?
N4	Distillation from LightGBM; teacher-guided hard negatives	Can the small network learn from the strong one?
N5	Focal, noise-robust (GCE, SCE) and multiple-instance losses; onset weighting	Do losses for rare, noisily labelled events help?
N6	Reliability-gated fusion, modality dropout, late fusion per modality	Does the detector survive missing sensors?
N7	Interpretability: which inputs do the LUTs read during true and false alarms?	Why does it fail, and what should change?

3.8 Evaluation Protocol and Metrics

Split. All design choices use five-fold cross-validation over the 96 training patients, split by patient, so a fold's validation patients are never seen in its training. The 29 test patients are scored once, at the end, with every choice frozen; any script that touches them needs an explicit flag and logs the access.

Event scoring (SzCORE). Following SzCORE [15], a detected event counts as correct if it overlaps a reference seizure, with 30 s of tolerance before onset and 60 s after the end; events closer than 90 s are merged. With TP correctly detected seizures, FN missed seizures and FP false alarms, we report per patient and then average over patients:

$$\text{Sens} = \frac{TP}{TP+FN}, \qquad \text{Prec} = \frac{TP}{TP+FP}, \qquad F_1 = \frac{2\,\text{Sens}\cdot\text{Prec}}{\text{Sens}+\text{Prec}}, \tag{3.3}$$

and the number of false alarms per 24 hours of recording. Window-level AUROC and AUPR are reported as well.

Decision logic. One decision is made per second; an alarm is raised when at least k of the last 10 decisions are positive, with an optional gate on the artefact flags. The threshold, k and the gate are chosen on the pooled out-of-fold predictions of the training patients, for two operating points: the best event-F1, and the highest sensitivity with at most one false alarm per hour.

Statistics. Methods are compared per patient with the paired Wilcoxon signed-rank test and a bootstrap 95 % confidence interval over patients. An extension is adopted only if $p < 0.05$ in its favour.

Hardware cost. We estimate the cost of each model analytically: comparators of the encoder, LUTs of the network, adders of the popcount and of the post-processing; multipliers are counted separately. Phase II replaces the estimates with synthesis results.

Paper metrics. For the comparison with the SeizeIT2 paper we also implement its own metrics (any-overlap sensitivity, false alarms per hour, AUROC, AUPR) exactly as in the released code, and report the result of correcting two defects of that code in a separate column.

3.9 Implementation and Computing Setup

The software is written in Python with PyTorch, scikit-learn and the gradient-boosting libraries; it comprises about 8,600 lines of code and 1,500 lines of unit tests (91 test functions) that run

automatically on every change on a synthetic dataset. It includes our own EDF reader that keeps the raw integer samples (bit-exact against pyEDFlib), the audit tools, the evaluation package (SzCORE scoring checked against the reference `timescoring` library and the paper's metrics ported from its code), the PyTorch port of ChronoNet (129,122 parameters, as the original model), the integer feature pipeline, all the models and extensions as configuration options, and the hardware handoff tools (export of a trained DWN, test vectors for every stage, a Verilog generator and an Icarus Verilog testbench). Every run writes its configuration, code version, log and metrics, and resumes after an interruption at fold and epoch level.

Experiments run on two machines: a 4-core CPU server for the feature extraction, the tree models and the small networks, and a laptop with an NVIDIA RTX 3060 GPU (6 GB) for ChronoNet, the large LUT networks and the raw-EEG models.

Chapter 4

Experimental Results and Comparisons

Sections 4.1 to 4.7 report the development results: five-fold patient-wise cross-validation on the 96 training patients, on which every design choice was made. Only after the design was frozen were the models scored, once, on the 29 held-out test patients (Section 4.8); the only earlier use of the test patients was the reproduction of the published ChronoNet baseline (Section 4.1), which the SeizeIT2 paper itself reports on them.

4.1 Reproduction of the Published Baseline

We re-ran the paper's deep-learning baseline, ChronoNet, on the 29 test patients with the official pre-processing. The released training code chooses its training epoch using these same test patients; we reproduced that protocol ("official") and also trained a "clean" version that uses the last epoch and no test information (Table 4.1).

Table 4.1 ChronoNet on the 29 test patients: published and reproduced (paper metrics).

	Seizures detected	**False alarms / h**	**AUROC**	**AUPR**
Published [9]	84.2 %	100.5	0.792	0.104
Ours, official protocol	74.5 %	22.6	0.741	0.109*
Ours, clean protocol	67.1 %	11.9	0.746	0.111*

* mean over the 90 test recordings that contain a seizure; pooled over all windows the AUPR is 0.004–0.005.

The published numbers are only partly reproduced: sensitivity is about 10 points lower, while false alarms are four to eight times lower. The AUPR matches when it is averaged per recording, which suggests that this is how the paper computed it; the paper's false-alarm rate matches neither of the two definitions in its code. We report the reproduction as measured, and the clean version is the ChronoNet result our models will be compared with at the final evaluation. Scored with SzCORE, the metric we use for our models, clean ChronoNet reaches a sensitivity of 0.79 at 106 false alarms per day, an event-F1 of 0.027.

4.2 Integer Versus Floating-Point Features

On 10 training patients, the integer features correlate with their floating-point twins at a median of 0.9965, the artefact flags agree on 99.7 % of windows, and a LightGBM classifier trained on either reaches the same leave-patients-out AUROC (0.736 integer against 0.735 floating point). The hardware arithmetic loses no measurable accuracy, so all further experiments use the integer features.

4.3 Comparison of Model Families

Table 4.2 compares nine model families at each model's best-F1 operating point, together with their estimated hardware cost, and Figure 4.1 plots event-F1 against estimated LUTs for every LUT-native run.

Table 4.2 Comparison of model families (five-fold cross-validation on the 96 training patients).

Model	AUROC	Event-F1	Sens.	FA / 24 h	Sens. at $\leq$1 FA/h	Estimated cost
LightGBM	0.870	**0.292**	0.327	3.6	0.594	≈30.6k LUTs (not a hardware design)
CatBoost	0.863	0.286	0.337	3.0	0.618	≈27.7k LUTs
XGBoost	0.867	0.277	0.286	3.4	0.603	≈47.9k LUTs
TreeLUT (5-bit leaves)	0.857	0.266	0.329	4.2	0.619	≈6.4k LUTs
DWN, 50 LUT6 (base)	0.840	**0.211**	0.315	**2.4**	0.581	**≈960 LUTs, no multipliers**
Random forest	0.852	0.198	0.216	0.5	0.536	≈323k LUTs
Logic-gate network	0.843	0.171	0.330	4.9	0.540	≈7.5k LUTs
INT8 MLP	0.830	0.147	0.225	7.8	0.520	9,600 multiplications
Logistic regression	0.806	0.105	0.216	8.8	0.409	117 multiplications

Three observations stand out:

- The DWN is by far the most compact model: about 6.6 times fewer LUTs than TreeLUT and 8 times fewer than the logic-gate network, with no multipliers, and one of the lowest false-alarm rates.
- It is still below the best tree models (against LightGBM the per-patient difference in event-F1 is significant, $p = 0.004$). Closing this gap is the aim of the extensions.

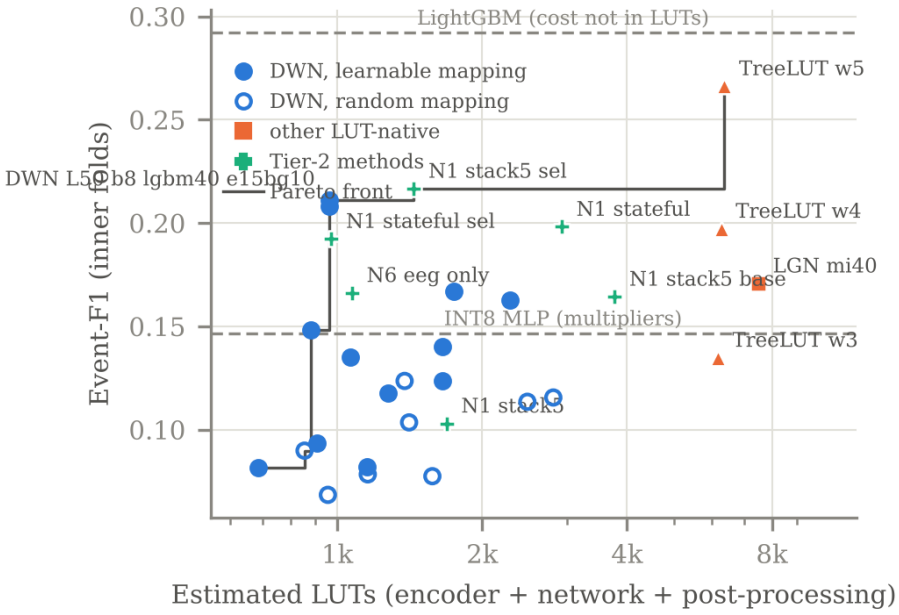


Figure 4.1 Event-F1 against estimated LUTs for the LUT-native models on the training folds, with the Pareto front; the best tree model and the INT8 MLP are shown as reference lines because their cost is not in LUTs.

- The INT8 MLP, the usual way to put a neural network on a chip, is worse than the DWN on every metric while needing 9,600 multiplications per decision.

4.4 Design of the LUT Network

Size, thresholds and mapping. A sweep over 50, 100 and 200 LUTs, 4, 8 and 16 thresholds per feature, and learned or random input mapping (17 of the 18 configurations finished; the last one is queued) showed no significant gain from size between 50 and 200 LUTs. Four thresholds were the worst at every size, and the learned mapping beat the random mapping in 7 of the 8 completed pairs. The smallest model that is not significantly worse than the best, 50 LUTs with 8 thresholds and learned mapping, became the base. Sizes of 500 to 2,000 LUTs are being trained on the GPU.

Feature selection. The interpretability study (Section 4.5) showed that mutual-information selection kept about 34 EEG features and missed the heart-rate and muscle features that the tree models rely on. Selecting the 40 inputs by LightGBM importance instead, inside each training set, improved the network at the same size (Table 4.3).

Training recipe. Training for 15 epochs with 10 background windows per seizure window gave the same result as 30 epochs with 20 (per-patient $p = 0.88$; event-F1 0.211, 2.4 false

Table 4.3 Effect of the feature selection on the 50-LUT DWN (per-patient comparison).

Feature selection	AUROC	Event-F1	FA / 24 h	Sens. at $\leq$1 FA/h	Est. LUTs
Mutual information (top 40)	0.819	0.148	7.0	0.476	883
LightGBM gain (top 40)	0.839	0.208	2.3	0.537	963

Per-patient event-F1 +0.060 [95 % CI +0.001, +0.119], $p = 0.047$; false alarms $p = 8 \times 10^{-7}$.

alarms per day) in a quarter of the time, so it is the base recipe for all further experiments.

4.5 First Results of the Extensions

Table 4.4 gives the extensions evaluated so far against the base DWN.

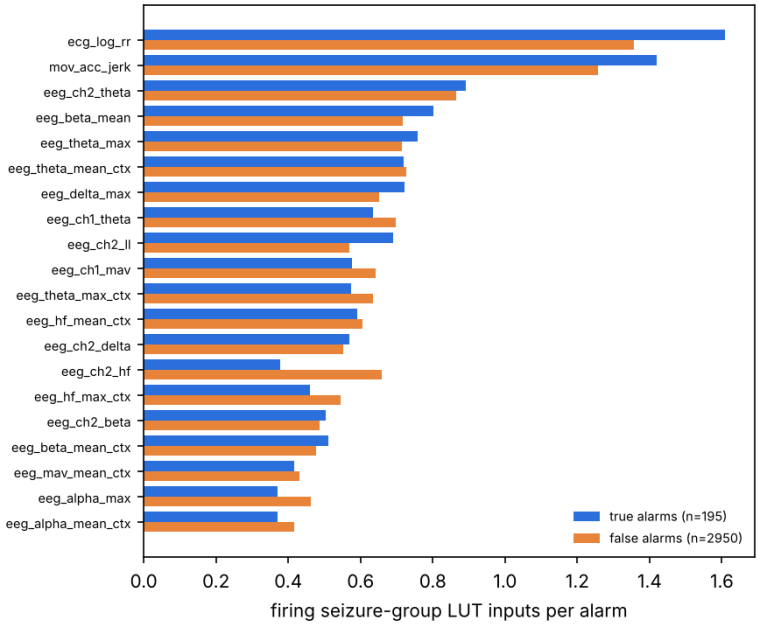


Figure 4.2 N7: inputs read by the firing seizure-group LUTs during true alarms ($n = 195$) and false alarms ($n = 2{,}950$), averaged per alarm, for the 50-LUT DWN with mutual-information features (out-of-fold predictions on the training patients).

4.6 Where the Detector Works and Where It Fails

Table 4.5 shows the sensitivity of the base DWN by seizure type, and Table 4.6 its behaviour when a sensor is removed at test time (out-of-fold predictions, best-F1 operating point).

The detector catches most tonic-clonic seizures, the dangerous ones, but few focal seizures, and the base network depends heavily on the heart signal and on motion. Training with random

Table 4.4 Extensions evaluated so far (per-patient comparison with the base DWN).

Extension	Result	Decision
N7 interpretability	During true and false alarms the network reads almost the same inputs: false alarms look like seizures to it. Its input usage agrees moderately with LightGBM's SHAP importance (Spearman 0.41). It revealed the feature-selection gap above.	Led to the new feature selection
N3 per-patient adaptation	Re-fitting the thresholds on each new patient's first 1, 2 or 4 hours raised the false alarms in every variant (for example from 6.8 to 10.2–39.5 per day at 1 h).	Not adopted
N1 stateful network (16 state bits)	Event-F1 0.192 against 0.211 ($p = 0.66$); false alarms 1.1 against 2.4 per day ($p = 0.013$); AUROC 0.861 against 0.840.	Not adopted alone; kept for the combination
N1 stacked windows (last 5 s)	Event-F1 0.217 against 0.211 ($p = 0.25$), about 480 more LUTs.	Not adopted
N5 loss functions (MIL, onset weighting, focal, GCE, SCE)	No event-F1 gain; the focal, GCE and SCE losses were significantly worse ($p = 0.038$, 0.005, 0.003).	Not adopted
N6 modality dropout	With ECG removed, event-F1 0.172 against 0.092 for the base ($p = 0.005$); with motion removed 0.192 against 0.136 ($p = 0.021$); with all sensors, 0.188 against 0.211 (not significant, $p = 0.23$).	**Adopted**
N6 late fusion	Event-F1 0.134 ($p = 0.0006$, worse) and collapses when any sensor is missing.	Dropped
N4 distillation from LightGBM	False alarms 0.9 against 2.4 per day ($p = 3 \times 10^{-7}$); event-F1 0.229 against 0.211 (not significant); 91 fewer estimated LUTs.	**Adopted (final network)**
N4 hard negatives	Teacher-guided hard negatives: event-F1 0.133 ($p = 0.0006$, worse); with distillation 0.215 (no gain).	Not adopted
N2 encoders	Uniform, Gaussian and learned thresholds: no event-F1 gain; uniform and strongly penalised learned thresholds significantly worse, although up to 759 LUTs cheaper.	Not adopted
N4 + N6 combined	Significantly worse than distillation alone with all sensors ($p = 0.030$), better without ECG ($p = 0.004$).	Not chosen; dropout kept as a variant

Table 4.5 Seizures detected by type.

Seizure type	Detected
FBTC	29 / 33 (88 %)
FIA	50 / 289 (17 %)
FA	26 / 274 (9 %)
Focal, awareness unknown	8 / 101 (8 %)

Table 4.6 Event-F1 with sensors removed (base 0.211).

Sensors removed	Event-F1
no ECG	0.092 ($p = 2 \times 10^{-5}$)
no motion	0.136 ($p = 0.004$)
no EMG	0.199 (n.s.)
EEG only	0.074 ($p = 2 \times 10^{-6}$)

modality dropout (N6, Table 4.4) removes most of this dependence: without ECG its event-F1 is 0.172 instead of 0.092.

4.7 Comparison with Published Research

Table 4.7 places our current results next to published results on behind-the-ear EEG. The comparison is indicative only: the published numbers are on other patient sets or use other metrics, and our models have not yet been evaluated on the test patients.

Table 4.7 Published results on behind-the-ear wearable EEG and our current results.

Method	Patients	Result	Comparable with ours?
ChronoNet, published [9]	SeizeIT2 test (29)	84.2 % sens., 100.5 FA/h (paper metrics)	No: test patients, other metric
ChronoNet, our clean run	SeizeIT2 test (29)	SzCORE: sens. 0.79, 106 FA/24 h, F1 0.027	Same metric, other patients
Patient-independent bte-EEG detector [7]	SeizeIT1	64.1 %, 2.8 FP/24 h	No: other dataset
Our DWN, 50 LUTs	SeizeIT2 training (96), cross-validation	SzCORE: sens. 0.315, 2.4 FA/24 h, F1 0.211; 0.581 at $\leq$1 FA/h	–

Section 4.8 gives the comparison on the same 29 test patients.

4.8 Final Evaluation on the Held-Out Test Patients

The design was frozen on the training patients (final network: the 50-LUT DWN with distillation from LightGBM). Every model was then retrained on all 96 training patients with five seeds and scored once on the 29 test patients, at the operating point chosen on the training folds; nothing was tuned on the test patients. Table 4.8 gives the result.

- The result held on unseen patients: the final network's event-F1 on the test patients (0.248) is above its cross-validation value (0.229), and the gain from distillation over the base network (0.172) is confirmed.
- It is significantly better than the clean ChronoNet baseline per patient ($p = 0.027$): ChronoNet detects more seizures (79 %) but raises over a hundred times more false alarms (106 against 0.8 per day).
- It is above TreeLUT at a seventh of the LUTs, above the INT8 network, level with random forests and close to CatBoost, and still below LightGBM ($p = 0.039$).

Table 4.8 Final evaluation on the 29 held-out test patients (SzCORE event metrics, mean over patients, mean ± std over 5 seeds; p: paired Wilcoxon on per-patient event-F1).

Model	Event-F1	Sens.	FA/24 h	p vs Chrono.	p vs LGBM	Estimated cost
DWN + distillation (final)	**0.248 ± 0.016**	0.303	**0.8**	0.027	0.039	**≈900 LUTs, 0 mult.**
DWN + modality dropout	0.242 ± 0.030	0.272	0.4	0.035	0.035	≈950 LUTs
DWN base	0.172 ± 0.044	0.340	2.7	0.011	0.004	≈980 LUTs
LightGBM	0.308 ± 0.013	0.331	0.6	0.027	–	≈30.8k LUTs
XGBoost	0.294 ± 0.014	0.301	0.4	0.062	0.29	≈48.3k LUTs
CatBoost	0.254 ± 0.009	0.330	1.2	0.031	0.026	≈27.7k LUTs
Random forest	0.246 ± 0.003	0.245	0.2	0.27	0.35	≈347k LUTs
TreeLUT	0.233 ± 0.044	0.292	1.0	0.016	0.005	≈6.3k LUTs
INT8 MLP	0.155 ± 0.008	0.249	1.7	0.017	0.005	9,600 multiplications
Logistic regression	0.123 ± 0.002	0.255	5.1	0.62	0.0008	117 multiplications
ChronoNet, clean (1 seed)	0.027	0.790	106.3	–	–	CNN + GRU

- At its other operating point (at most one false alarm per hour) it detects 54 % of the test seizures.

4.9 Hardware Handoff Check

The final network was exported as its thresholds, input mapping, 50 truth tables and popcount head, and test vectors were generated for every stage, from the raw integer samples through features, thermometer bits, LUT outputs and score to the alarms, on 10 test recordings. The generated reference Verilog, simulated in Icarus Verilog, produced exactly the same outputs as the software on **all 431,960 one-second windows** of these recordings. The FPGA implementation in Phase II therefore starts from a verified, bit-exact description.

Chapter 5

Challenges Faced and How They Were Addressed

This chapter describes the main problems met in Phase I and how each was solved. Every decision is recorded with its evidence in the project's decision log, including the ideas that did not work.

5.1 Data

Table 5.1 Challenges with the data.

Challenge	What happened	How we addressed it
The data did not match its documentation	The paper says 250 Hz and 886 seizures; the files say 256 Hz and 883. Centre and age labels, needed for stratified splits, are not in the public release; seizure-visibility labels are empty.	A full audit of all 11,009 files before any modelling. Every window and filter is defined in seconds and designed for 256 Hz; folds are stratified by seizure count; analyses the data cannot support were dropped and reported.
126 GB of data on a machine without the disk or a GPU	The dataset could not be stored where the code runs.	The audit and the feature extraction stream each file from the public S3 bucket, process it and delete it; only the 3.5 GB of integer features are kept.

Challenge	What happened	How we addressed it
Silent download truncation	Comparing integer and floating-point features exposed one ECG file cut at 51,855 of 71,835 s. A new verifier then found 9 recordings silently truncated by incomplete network reads (one EEG file had 12 of 16,363 windows).	Downloads are now checked against the expected size and every cached recording against the audit durations. All 9 were re-extracted, and the 3 model runs trained on the bad cache were repeated.
A recording with an ECG lead but no detectable heartbeat	The feature extractor crashed on an empty list of beats (3,250 s of data).	The recording is kept, its ECG features are set to 0 and a quality bit is raised.
The official artefact rule does not suit behind-the-ear EEG	The 13 µV RMS floor flags 69 % of ordinary windows on one channel, because behind-the-ear amplitudes are close to it.	For our models the artefact flags are input bits, and whether to gate alarms on them is chosen on the training folds.

5.2 Reproducing the Published Baseline

Table 5.2 Challenges in reproducing the published baseline.

Challenge	What happened	How we addressed it
ChronoNet did not reproduce	Our run detected 74.5 % of seizures against 84.2 % published, with 22.6 instead of 100.5 false alarms per hour.	We ported the official code literally (identical outputs on 900 random test cases) and then checked the metric definitions: the published AUPR matches only when averaged per recording (0.109 against 0.104), and the published false-alarm rate matches neither definition in the code. The gap is reported as measured.
The released code selects its model on the test patients	Its training loop keeps the epoch with the best score on the same patients it is then evaluated on.	We reproduced it as published and also trained a clean version (last epoch, no test information), which is the honest reference for the final comparison.

Challenge	What happened	How we addressed it
Defects in the released evaluation code	True positives can be counted more than once, and the artefact gate (`a and b` on two Python lists) silently uses only one channel.	Reproduced as written for the comparison with the paper; corrected numbers reported in a separate column.
The SVM baseline cannot be rebuilt yet	The dataset paper defers its 42 features to Vandecasteele et al. [7], whose full text was not available to us.	Still open: we need that paper's feature table.

5.3 Hardware Arithmetic

Table 5.3 Challenges with the integer (hardware) arithmetic.

Challenge	What happened	How we addressed it
Rounding noise in integer filters	With plain rounding, the 0.5 Hz high-pass filter had an error of 140 LSB standard deviation, larger than the signal itself (61 LSB).	Second-order error feedback of the discarded bits reduced the error to 0.7 LSB.
Quantised coefficients moved the filter zeros	Rounding each coefficient independently moved the low-pass zeros away from the Nyquist frequency (stop-band floor -55 dB).	Structure-preserving quantisation, $b = b_0\,(1, \pm 2, 1)$, with coefficients as sums of powers of two.
Heart rate without a divider	Heart rate needs $60/\mathrm{RR}$.	We use $\log_2(\mathrm{RR})$ from integer R-peak timestamps: no division and no multiplication.
Proving that the integer pipeline costs nothing	Hardware arithmetic could lose accuracy silently.	A floating-point twin of every feature: median correlation 0.9965 and equal LightGBM AUROC (0.736 against 0.735).
TreeLUT scores were not comparable across folds	Each fold's integer score had its own scale, so pooled thresholds were meaningless; 3-bit leaves could not represent the model.	Scores converted to a common unit; leaf width chosen on the folds (5 bits).

Challenge	What happened	How we addressed it
Verilog disagreed with the software	Thresholds outside a feature's bit width (for example 79 on a 2-bit presence feature) produced wrong comparator logic.	The Verilog generator now emits constant bits for such thresholds; the simulation agrees with the software on 100 % of the test vectors.

5.4 Computing and Infrastructure

Table 5.4 Challenges with computing resources.

Challenge	What happened	How we addressed it
Limited GPU resources	Our GPU is a laptop RTX 3060 with 6 GB of memory and 7.4 GB of system memory, and the reference DWN library runs only on CUDA.	We re-implemented the DWN in PyTorch, identical to the library's CUDA kernel (unit-tested), so that it also runs on CPUs. CPU jobs run on a 4-core server; large networks and raw-EEG models run on the GPU.
Long runs interrupted	Machine restarts stopped long runs; the logic-gate network lost 3 of its 5 folds once.	Every run resumes at fold and epoch level, and a queue skips finished runs; a resumed run was checked to give identical metrics.
Out-of-memory failures	The stateful network's sequence sampler held 24 million windows in memory (13.9 GB of a 15 GB machine).	A new sampler reads each recording once and stores bits as bytes: the same sequences with a peak of 3.1 GB.
Two machines, one code base	The CPU server and the GPU laptop must not overwrite each other's work.	One version-controlled code base, written run instructions for each machine, and automatic tests on every change.

5.5 Machine Learning

Table 5.5 Challenges in the machine-learning design.

Challenge	What happened	How we addressed it
Seizures are 0.123 % of the time	Accuracy is meaningless, and training on all windows teaches the model never to fire.	Event-based scoring (SzCORE) per patient; training on all seizure windows plus a fixed ratio of background windows; operating points chosen for a false-alarm budget.
Results must hold on unseen patients	Mixing a patient's windows between training and validation inflates results.	Patient-wise five-fold cross-validation; feature selection and thresholds fitted inside each training fold only; the test patients locked behind a flag, with every access logged.
Early results were weaker than expected	The first LUT network, with mutual-information features, reached an event-F1 of 0.148 with 7.0 false alarms per day; the interpretability study showed that it ignored the heart-rate and muscle features.	Feature selection by LightGBM importance: event-F1 0.208 and 2.3 false alarms per day at the same size.
A confounded comparison	The stateful network had received all 117 features instead of the selected 40, so its apparent gain mixed two changes.	Found by checking the run's records; all models now share one training function, and the run was repeated.
Stacking past windows picked duplicate features	Ranking the 585 stacked columns kept lagged copies of only 11–13 distinct features.	Rank on the current window, then keep every lag of the selected features.

Chapter 6

Discussion, Conclusions and Future Scope

6.1 Discussion and Current Findings

Accuracy per LUT. The 50-LUT DWN gives the best accuracy per unit of hardware among the models compared: an event-F1 of 0.211 with 2.4 false alarms per day at about 960 estimated LUTs and no multipliers, against 0.266 for TreeLUT at about 6.4k LUTs and 0.292 for LightGBM, which is not a hardware design. The INT8 MLP, the conventional way to put a neural network on a chip, is worse on every metric while needing 9,600 multiplications per decision. Within the DWN family, size did not matter between 50 and 200 LUTs, while the choice of input features and the encoder resolution did.

The encoder dominates the cost. About 850 of the 960 estimated LUTs are comparators of the thermometer encoder; the network itself is 50 LUTs. This agrees with recent measurements of DWN hardware [58] and makes the encoder (extension N2) the main lever for reducing the hardware further.

False alarms look like seizures to the network. The interpretability study showed that the network reads almost the same inputs during true and false alarms. It also exposed a weakness of the first feature selection, which ignored the heart signal; fixing it was the largest single improvement of Phase I. Inspecting the trained LUTs was therefore more useful than tuning the network blindly.

Tonic-clonic against focal seizures. The detector finds 88 % of tonic-clonic seizures but 8–17 % of focal seizures, and loses most of its performance without ECG. This matches the literature, where wearable detection is established only for tonic-clonic seizures [5], and it shows where the remaining work lies: focal seizures and robustness to missing sensors.

Reproducibility. The published ChronoNet baseline could only be partly reproduced, and its released code selects the model on the test patients. Keeping our own test patients untouched until the design is frozen, and reporting a clean version of the baseline, is therefore essential for an honest final comparison.

6.2 Current Progress and Future Scope

Table 6.1 summarises the work carried out in the two months of Phase I, and Table 6.2 the status of every activity at the end of Phase I.

Table 6.1 Work carried out during Phase I.

Period	Activities
Month 1 (September 2026)	Selection of the topic and definition of the problem with the guide; literature study on wearable seizure detection and LUT-based neural networks; study of the SeizeIT2 dataset, its baselines and its evaluation; setting up the software tools and data access; first experiments, whose results were weaker than those reported here.
Month 2 (October – November 2026)	Complete audit of the dataset; reproduction of the ChronoNet baseline (official and clean); integer feature pipeline and fixed-point specification; cross-validated comparison of nine model families; design study of the DWN and the improved feature selection; extensions N1–N7 and the frozen final design; the single evaluation on the 29 held-out test patients; hardware handoff with the Verilog check; this report.

Table 6.2 Status of the project activities at the end of Phase I.

Activity	Status	Remarks
Literature review	Completed	79 references in five areas; research gaps identified
Dataset access and audit	Completed	All 11,009 files measured; differences from the documentation recorded
Reproduction of ChronoNet	Completed	Partly reproduced; the clean run is the reference for the final comparison
Reproduction of the SVM baseline	Pending	Needs the 42-feature definition of the original study
Integer feature pipeline and fixed-point specification	Completed	117 features; no measurable accuracy loss
Comparison of model families	Completed for nine families	LogicNets-style network, GRU and raw-EEG CNN/TCN running

Activity	Status	Remarks
DWN design study	Completed up to 200 LUTs	500–2,000 LUTs and further ablations running on the GPU
Extensions N1, N3, N7	Completed	None adopted alone; N7 led to the improved feature selection
Extensions N5, N6	Completed	Modality dropout (N6) adopted; no loss variant (N5) adopted
Extensions N2, N4	Completed	Distillation (N4) adopted for the final network; encoders and hard negatives not adopted
Final evaluation on the 29 test patients	Completed	10 models $\times$ 5 seeds; final network event-F1 0.248, 0.8 false alarms per day; the logic-gate network's run finishes separately
Hardware handoff (export, test vectors, reference Verilog)	Completed	Verilog matches the software on all 431,960 windows of 10 test recordings
FPGA implementation	Planned	Synthesis, measurements and demonstration in Phase II

Phase II will take the verified design to hardware: the final network will be synthesised on a low-power FPGA and its LUTs, power and latency measured against the INT8 MLP and TreeLUT on the same device, and the recordings streamed through the board. In parallel, the large LUT networks and the raw-EEG references will complete the model comparison, and the open questions of Phase I (focal seizures, the dependence on ECG) will be studied on the training patients. Table 6.3 gives the month-wise plan.

Table 6.3 Month-wise plan for the remaining project period.

Month	Work	Expected deliverable
December 2026	FPGA tool flow; synthesis and place-and-route of the verified design; large LUT networks and raw-EEG references on the training patients	Resource and timing report
January 2027	Conference paper (EMBC 2027, full papers reportedly due 24 January 2027); power and latency measurement	Submitted paper
February 2027	Measured LUTs, power and latency; INT8 MLP and TreeLUT on the same device	Hardware comparison table
March 2027	On-board demonstration with recordings streamed through the design; documentation	Demonstration
April 2027	Final report and presentation	Final report

6.3 Conclusions

Phase I established the foundation of the project and its first results:

1. The SeizeIT2 dataset was audited completely; its sampling rate, seizure count and labels differ from the documentation, and the design follows the files.

2. The published ChronoNet baseline was reproduced only partly, and a clean version without test information was trained as the reference for the final comparison.

3. A bit-exact integer feature pipeline for behind-the-ear EEG, ECG, EMG and motion loses no measurable accuracy against floating point.

4. A 50-LUT Differentiable Weightless Neural Network reaches an event-F1 of 0.211 with 2.4 false alarms per day on cross-validation, at about 960 estimated LUTs and no multipliers: seven to eight times smaller than the other LUT models and better than an INT8 network, though still below gradient-boosted trees.

5. Interpretability of the LUT network led to the largest improvement of the phase, and the thermometer encoder was identified as the dominant hardware cost.

6. Training with random modality dropout makes the network robust to a missing ECG or motion sensor (event-F1 0.172 against 0.092 without ECG) at no significant cost when all sensors are present.

7. Distillation from LightGBM was adopted for the final network. On the 29 held-out test patients, scored once with five seeds, it reaches an event-F1 of 0.248 with 0.8 false alarms per day at about 900 LUTs: significantly better than the clean ChronoNet baseline, above TreeLUT and the INT8 network, level with random forests and below LightGBM.

8. The generated Verilog of the final network matches the software on all 431,960 windows of 10 test recordings.

9. No LUT-native seizure detector exists in the literature, and no LUT network fuses modalities or handles missing sensors; the project addresses both.

Phase II will implement and measure the detector on an FPGA and continue the work on focal seizures and on robustness to missing sensors.

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Appendix A

Sources of the Reported Numbers

Every number in this report comes from a saved result file of the project. Table A.1 lists where each group of numbers is stored; the run folders hold the configuration, the code version, the log and the metrics of each experiment.

Table A.1 Sources of the reported numbers.

Numbers	Source
Dataset inventory, seizure classes, differences from the documentation, signal quality (Chapter 3)	`reports/data_audit.md`, `reports/data_audit.json`, `results/audit/`
Fixed-point specification of the features	`spec/fixed_point.md`
Integer against floating-point features	`results/m3/int_float_gap.md`
ChronoNet reproduction (Table 4.1)	runs `20261107-125159_chrononet_official_s1` and `20261107-204035_chrononet_clean_s1`
Comparison of model families (Table 4.2)	`results/tables/inner_cv.md` and the run folders named there
Effect of the feature selection (Table 4.3)	runs `dwn_L50_b8_learn` and `dwn_L50_b8_lgbm40`
Base DWN, training recipe, seizure types and missing sensors	run `20261108-082813_dwn_L50_b8_lgbm40_e15bg10_s0` (`metrics.json`, `breakdown.md`)

Numbers	Source
Extensions (Table 4.4)	`results/tables/m5_ablation.md`; `n7_interpretability.json` and `n3_adaptation.json` in the run folder of `dwn_L50_b8_learn`
Decisions and their evidence	`results/decisions.md`